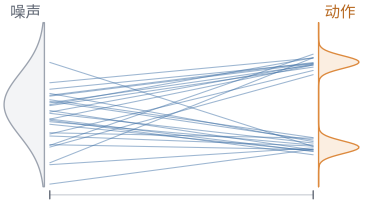
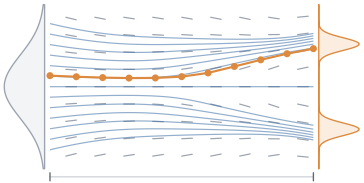
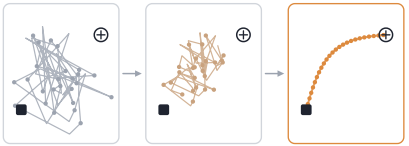
(a) 流匹配：从噪声到动作作块



训练：随机（噪声，动作）配对之间的直线路径

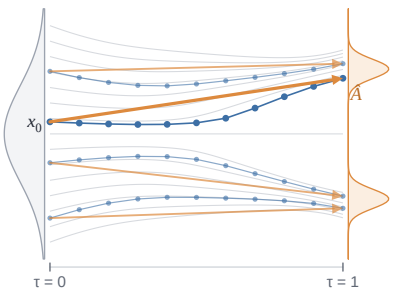


采样：沿学到的速度场积分，N = 10 步（圆点）



一个动作块的二维夹爪路径（■ 起点，⊕ 目标）

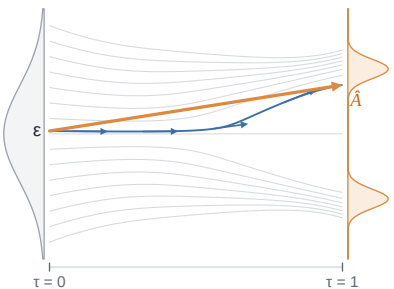
(b) 蒸馏：N 步 → 1 步



$$L(\varphi) = \|S_\varphi(x_0, o) - \Phi^{(N)}_\theta(x_0, o)\|^2$$

NFE：N → 1，需要训练好的教师

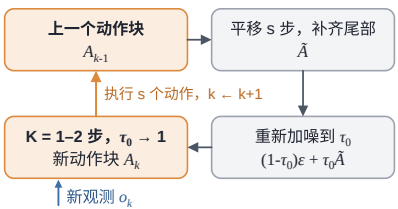
(c) 一步训练目标：平均速度



一步生成：$\hat{A} = \varepsilon + u_\theta(\varepsilon, 0, 1 \mid o)$

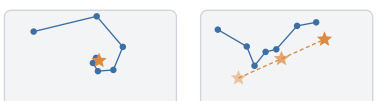
直接学习平均速度：无需教师，NFE = 1

(d) 热启动与跨步迭代



$$\|T_o(A) - T_o(A')\| \le \rho\|A - A'\|, \quad \rho < 1$$

仅在观测 o 固定时成立



o 固定：收敛到 $A^*(o)$　　o_k 变化：$A^*(o_k)$ 漂移